Day 4: Diffusion Models — From Prompts to Images and Video

Study notes • Diffusion fundamentals, image generation, prompt design, and video generation

Note: These are newly prepared study notes aligned with your day-wise schedule, not pages extracted from the original AMD PDF.

1. Learning objectives

• Describe how diffusion models learn to reverse a gradual noising process.

• Understand text conditioning, denoising, and decoding in image-generation pipelines.

• Explain why video generation must maintain both frame quality and temporal consistency.

2. What is a diffusion model?

• A diffusion model learns to generate data by reversing a process that gradually adds noise to training examples.

• Forward process: progressively add noise to an image until it becomes noise-like. Reverse process: start from noise and repeatedly denoise with a trained neural network.

• During training, the model learns to predict the noise or another denoising target at different timesteps. Exact training objectives and samplers vary by model.

• Many modern systems use latent diffusion: they operate in a compressed representation rather than directly on every image pixel, reducing computational cost.

3. Text-to-image pipeline

• Stage | Purpose

• Prompt encoding | Convert text into a representation used for conditioning.

• Noise initialization | Create a starting noisy image or latent.

• Denoising | Iteratively predict and remove noise, guided by conditioning.

• Decoding | Convert a final latent into an image when using latent diffusion.

• Post-processing | Optional resizing, upscaling, editing, and safety checks.

4. Writing effective image prompts

• A useful structure is: subject + action or pose + environment + composition + lighting + visual style + palette + framing + constraints.

• Example: “A small research robot examining plants in a greenhouse, wide composition, soft morning light, realistic materials, clear foreground and background, natural green palette.”

• Specify relationships and layout: foreground versus background, camera angle, object placement, and what must remain visible.

• Avoid contradictory instructions. Put the most important subject and action near the start of the prompt.

5. Generation controls

• Guidance strength affects how strongly a system follows its conditioning when that control is available; excessive guidance can create artifacts.

• Sampling steps affect latency and quality, but more steps do not guarantee a better image.

• A seed can help reproduce a result when the model and settings are unchanged.

• Aspect ratio and resolution influence composition, compute, and memory use.

• Negative prompts can describe unwanted elements in systems that support them; effectiveness varies by model.

6. From images to video

• Text-to-video systems generate sequences of frames that should look plausible individually and remain consistent over time.

• A video prompt should specify subject, setting, action, camera movement, temporal progression, continuity constraints, and output settings when available.

• Example structure: “A cyclist rides along a tree-lined path at sunrise. The camera tracks smoothly from the side; leaves move gently in the breeze; lighting and the cyclist’s appearance remain consistent.”

• Temporal consistency means identities and objects should not flicker or change unexpectedly. Motion coherence means actions should follow plausible paths and interactions.

7. Challenges and quality review

• Video generation is computationally expensive because many frames must be generated and coordinated.

• Longer sequences increase the challenge of maintaining identities, objects, lighting, and story continuity.

• Evaluate prompt alignment, visual quality, motion, consistency, and artifacts—not just whether a single frame looks good.

• Inspect outputs for distorted objects, unwanted text, inconsistent geometry, and scene changes that were not requested.

• Generated imagery should not be treated as factual evidence of a real event. Respect privacy, consent, copyright, and applicable rules.

Practical Exercises

1. Write three prompts for one subject in photorealistic, illustration, and cinematic styles.

2. Design a five-second video prompt with a subject, action, camera movement, and clear beginning-to-end sequence.

3. Evaluate two generated images for prompt alignment, composition, artifacts, and consistency.

Review Questions

1. What is the forward diffusion process?

2. What does a denoising network learn to predict?

3. Why do many systems use a latent representation?

4. What elements make an image prompt effective?

5. Why is temporal consistency harder in video than in a single image?

6. What are the limitations of seeds and negative prompts?